# COURSE NAME: COMPUTER NETWORKS
# COURSE CODE: BCS502
# Module 1

## 1.1 DATA COMMUNICATIONS

Data communications are the exchange of data between two devices via some form of transmission medium such as a wire cable. For data communications to occur, the communicating devices must be part of a communication system made up of a combination of hardware (physical equipment) and software (programs). The effectiveness of a data communications system depends on four fundamental characteristics: delivery, accuracy, timeliness, and jitter.

**1. Delivery.** The system must deliver data to the correct destination. Data must be received by the intended device or user and only by that device or user.

**2. Accuracy.** The system must deliver the data accurately. Data that have been altered in transmission and left uncorrected are unusable.

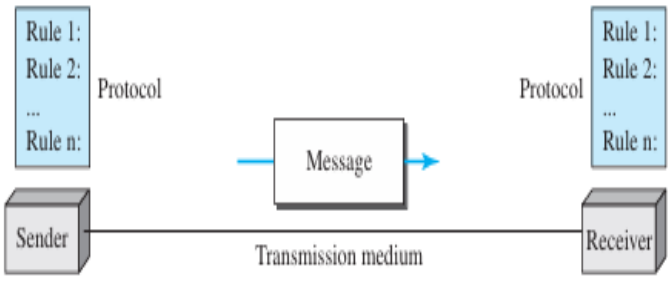
**3. Timeliness**. The system must deliver data in a timely manner. Data delivered late are useless. In the case of video and audio, timely delivery means delivering data as they are produced, in the same order that they are produced, and without significant delay. This kind of delivery is called real-time transmission.

**4. Jitter**. Jitter refers to the variation in the packet arrival time. It is the uneven delay in the delivery of audio or video packets. For example, let us assume that video packets are sent every 30 ms. If some of the packets arrive with 30-ms delay and others with 40-ms delay, an uneven quality in the video is the result.

### 1.1.1 Components

A data communications system has five components (see Figure 1.1).

Figure 1.1 *Five components of data communication*



**1. Message**. The message is the information (data) to be communicated. Popular forms of information include text, numbers, pictures, audio, and video.
**2. Sender.** The sender is the device that sends the data message. It can be a computer, workstation, telephone handset, video camera, and so on.
**3. Receiver.** The receiver is the device that receives the message. It can be a computer, workstation, telephone handset, television, and so on.
**4. Transmission medium**. The transmission medium is the physical path by which a message travels from sender to receiver. Some examples of transmission media include twisted-pair wire, coaxial cable, fiber-optic cable, and radio waves.
**5. Protocol.** A protocol is a set of rules that govern data communications. It represents an agreement between the communicating devices. Without a protocol, two devices may be connected but not communicating, just as a person speaking French cannot be understood by a person who speaks only Japanese.

### 1.1.2 Data Representation

Information today comes in different forms such as text, numbers, images, audio, and video.

Data representation refers to the method of converting different types of information into **binary form (0s and 1s)** so that computers can store, process, and transmit data.

**1. Text**

- Text is represented in computers as a **sequence of bits**.
- Each character is assigned a **unique binary code**.
- This process is called **coding**.
- The most widely used coding system is **Unicode**, which uses **32 bits** to represent characters from all languages.
- **ASCII** is an older coding system and forms the **first 127 characters of Unicode**.

**2. Numbers**

- Numbers are also represented using **binary values**.
- Unlike text, numbers are **directly converted into binary**.
- This makes **mathematical calculations easier and faster**.
- Different number systems such as **binary, decimal, and hexadecimal** are used.

**3. Images**

- Images are represented as a **matrix of pixels**.
- Each pixel is assigned a **binary value**.
- Higher number of pixels gives **better resolution** but needs **more memory**.
- Black and white images use **1 bit per pixel**.
- Gray-scale images use **more bits** (e.g., 2 bits for 4 gray levels).
- Color images are commonly represented using the **RGB model (Red, Green, Blue)**.
- Another color model used is **YCM (Yellow, Cyan, Magenta)**.

**4. Audio**

- Audio represents **sound or music**.
- Audio signals are **continuous in nature**.
- Sound is converted into an **electrical signal using a microphone**.
- Audio data representation is discussed in **Chapter 26**.

**5. Video**

- Video refers to **moving pictures or movies**.
- Video can be:
    - **Continuous** (captured by a video camera), or
    - **Discrete**, made of a sequence of images (frames).
- The rapid display of images creates the **illusion of motion**.
- Video representation is also discussed in **Chapter 26**.

**1.1.3 Data Flow**

Communication between two devices can be simplex, half-duplex, or full-duplex as shown in Figure 1.2.

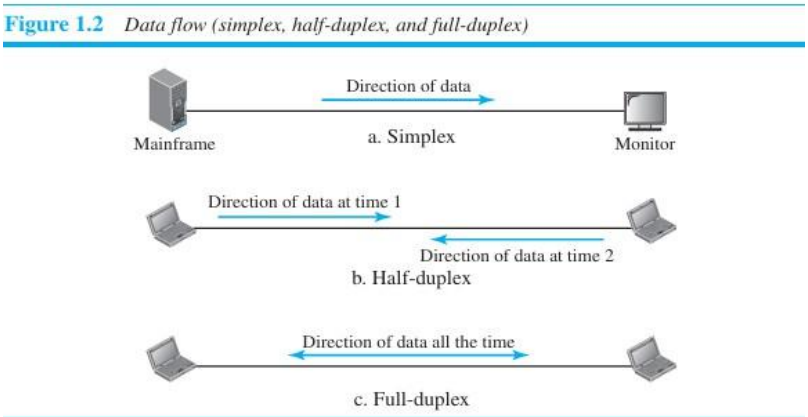



**Figure 1.2** *Data flow (simplex, half-duplex, and full-duplex)*

**Modes of Data Transmission**

**1. Simplex Mode**

- In simplex mode, communication is **one-way only**.
- Only **one device transmits**, and the other **only receives**.
- The receiver **cannot send data back**.
- The **entire channel capacity** is used in one direction.
- **Examples:** Keyboard (input only), Monitor (output only), TV broadcast.

**2. Half-Duplex Mode**

- In half-duplex mode, **both devices can transmit and receive**, but **not at the same time**.
- When one device is sending, the other **only receives**.
- Communication happens **one direction at a time**.
- The **full capacity of the channel** is used by the transmitting device.
- **Examples:** Walkie-talkies, CB radios.

**3. Full-Duplex Mode**

- In full-duplex mode, **both devices can transmit and receive simultaneously**.
- Data flows in **both directions at the same time**.
- The channel capacity is **shared between both directions**.
- This can be achieved using:
    - Two separate communication paths, or
    - Dividing the channel capacity.
- **Example:** Telephone communication.

## 1.2 NETWORKS

A **network** is an interconnection of two or more devices that can **communicate and share data** with each other.

- Devices in a network can be:
    - **End devices (hosts):** Computer, laptop, mobile phone, workstation, security system, etc.
    - **Connecting devices:**
        - **Router:** Connects one network to another
        - **Switch:** Connects multiple devices within a network
        - **Modem:** Converts data into a form suitable for transmission
- Devices are connected using **wired media** (cables) or **wireless media** (air).
- Even a small home setup connecting two computers using a router is considered a **network**.

### 1.2.1 Network Criteria

A good network must satisfy certain important criteria. The main criteria are:

**1. Performance**

- Performance refers to **how fast and efficiently** a network operates.
- It is measured using:
    - **Transit Time:** Time taken for data to travel from sender to receiver.
    - **Response Time:** Time between a request and its reply.
- Network performance depends on:
    - Number of users
    - Type of transmission medium
    - Hardware capability
    - Software efficiency
- Performance metrics include:
    - **Throughput:** Amount of data transferred per unit time
    - **Delay:** Time taken for data to reach the destination
- Higher throughput often causes **more delay** due to network congestion.

**2. Reliability**

- Reliability refers to the **ability of the network to function correctly**.
- It is measured by:
    - Frequency of failures
    - Time taken to recover from failures
    - Ability to operate during disasters
- A reliable network ensures **accurate and continuous data delivery**.

**3. Security**

- Security protects network data from **unauthorized access**.
- It ensures:
    - Protection of data from damage and modification
    - Prevention of data theft
    - Safe recovery from security breaches and data loss
- Security is achieved through **policies, procedures, and protection mechanisms**

### 1.2.2 Physical Structures

**Types of Connection**

A network **consists** of two or more devices connected through **links**. A link **is** a communication path that transfers data between devices. For communication to occur, devices must be connected to the same link at the same time.

There are **two types of connections**:

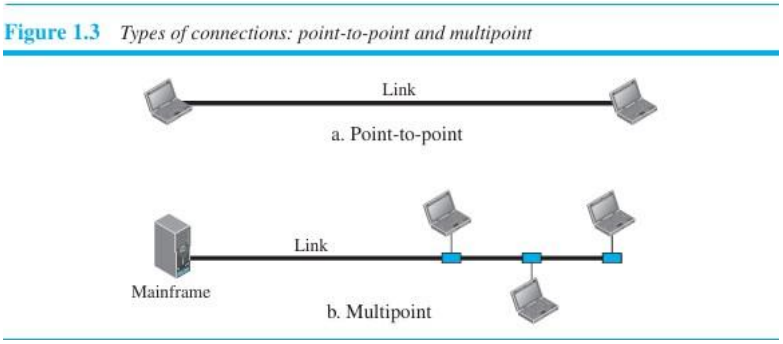
**1. Point-to-Point Connection**

- A **point-to-point** connection provides a **dedicated link between two devices**.
- The **entire capacity of the link** is used only by these two devices.
- The connection can be wired (cable) or wireless (microwave, satellite).
- Communication occurs **directly between the two devices**.
- **Example:**
    - Infrared remote control and television
    - Direct cable connection between two computers

**2. Multipoint (Multidrop) Connection**

- A **multipoint connection** is shared by **more than two devices**.

- The channel capacity is **shared among devices**.
- Sharing can be:
    - **Spatial sharing:** Multiple devices transmit at the same time
    - **Time sharing:** Devices take turns to transmit
- Used when many devices need to communicate using a single link.




**Physical Topology**

**Physical topology** refers to the **physical layout of devices and links** in a network. It shows how devices (nodes) are **connected to each other**.

There are **four basic physical topologies**:

1. Mesh
2. Star
3. Bus
4. Ring

**1. Mesh Topology**

- In a **mesh topology**, every device is connected to **every other device**.
- Each connection is a **dedicated point-to-point link**.
- Number of links required = $n(n-1)/2$, where $n$ is the number of devices.
- Each device requires **n − 1 I/O ports**.

**Advantages:**

- No traffic congestion due to dedicated links
- Highly **reliable and robust**
- High **security and privacy**
- Easy **fault detection and isolation**

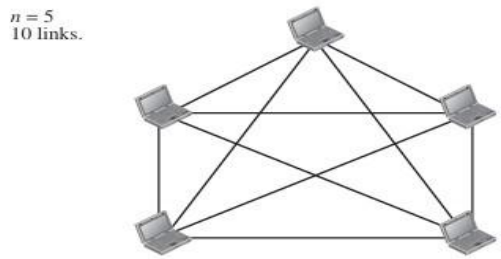
**Disadvantages:**

- Requires **large amount of cabling**
- Very **expensive**
- Difficult to install and maintain

**Example:**

- Telephone regional office networks

**Figure 1.4** *A fully connected mesh topology (five devices)*




**2. Star Topology**

- All devices are connected to a **central device** called a **hub or switch**.
- Devices are **not directly connected to each other**.
- All communication passes through the hub.

**Advantages:**

- Easy to install and reconfigure
- Less cabling than mesh
- Easy fault detection
- Failure of one link does not affect others

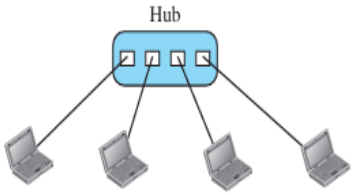
**Disadvantages:**

- **Hub failure** causes complete network failure
- Requires more cabling than bus or ring in some cases

**Example:**

- Local Area Networks (LANs)

**Figure 1.5** *A star topology connecting four stations*




**3. Bus Topology**

- Uses a **single long cable (backbone)** to connect all devices.
- Devices connect using **drop lines and taps**.
- It is a **multipoint topology**.

**Advantages:**

- Easy to install
- Uses **less cable**
- Cost-effective

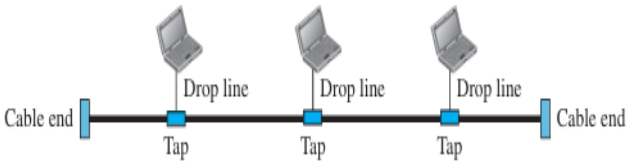
**Disadvantages:**

- Difficult fault isolation
- Break in backbone stops entire network
- Signal strength decreases with distance
- Adding devices is difficult

**Example:**

- Early Ethernet LANs

Figure 1.6 *A bus topology connecting three stations*




**4. Ring Topology**

- Each device is connected to **two neighboring devices**, forming a ring.
- Data travels in **one direction**.
- Each device contains a **repeater** to regenerate signals.

**Advantages:**

- Easy to install and reconfigure
- Simple fault detection
- Equal access for all devices

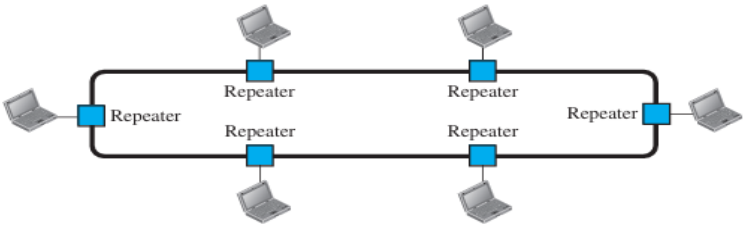
**Disadvantages:**

- Failure of one device can break the network
- Unidirectional data flow causes delay
- Less suitable for high-speed networks

**Example:**

- IBM Token Ring network

**Figure 1.7** *A ring topology connecting six stations*




## 1.3 Network Types

Networks can be classified based on **size, geographical coverage, and ownership**. The most common types of networks are **LAN, WAN**, and **Internetworks**. Switching is used to connect these networks.

### 1.3.1 Local Area Network (LAN)

- A **LAN** is a privately-owned network.

- It connects devices within a **small geographical area** such as:
    - Home
    - Office
    - Building
    - Campus
- A LAN can range from **two computers and a printer** to a large corporate network.
- Each device (host) in a LAN has a **unique address**.
- Data packets contain:
    - Source address
    - Destination address

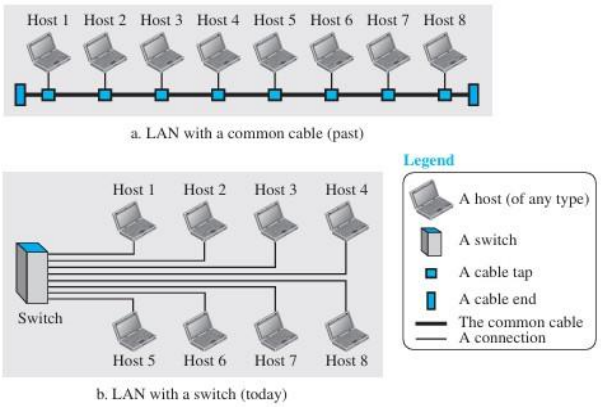
**Traditional LAN**

- All devices were connected using a **common cable**.
- Packets were received by **all devices**, but only the destination accepted it.

**Modern LAN**

- Uses a **switch**.
- The switch forwards packets **only to the destination device**.
- Reduces traffic and allows **simultaneous communication** between multiple pairs.
- LANs mainly support **resource sharing**.
- LANs are usually connected to other LANs or WANs today.

**Figure 1.8** *An isolated LAN in the past and today*




### 1.3.2 Wide Area Network (WAN)

- A **WAN** covers a **large geographical area** such as:
    - City
    - State
    - Country
    - World
- A LAN connects **hosts**, whereas a WAN connects **connecting devices** such as:
    - Routers
    - Switches
    - Modems
- LANs are usually **privately owned**.
- WANs are usually **owned and operated by service providers** and leased to users.

**Types of WANs**

**1. Point-to-Point WAN**

- Connects **two devices** directly.
- Uses a dedicated transmission medium.
- Commonly used to connect networks together.

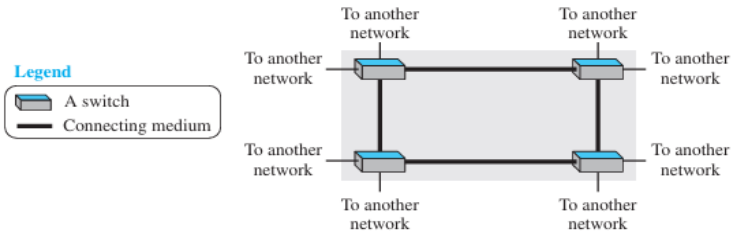
**2. Switched WAN**

- Connects **more than two devices**.
- Consists of several point-to-point links connected by switches.
- Forms the **backbone of global communication networks**.

**Figure 1.9** *A point-to-point WAN*




**Figure 1.10** *A switched WAN*

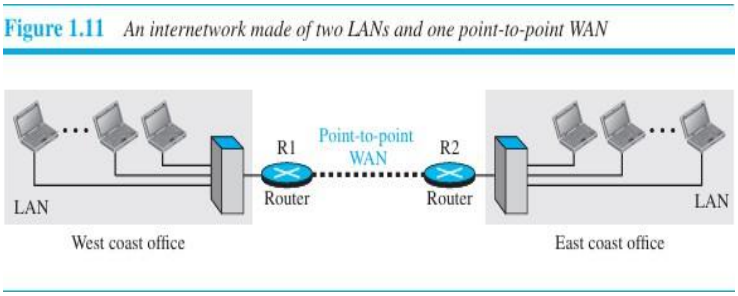



**Internetwork**

- An **internetwork (internet)** is formed by connecting **two or more networks**.
- Example:
  - Two offices with separate LANs connected using a WAN.
- Routers are used to:

- Block local traffic
- Forward data between networks

- Enables communication across different geographical locations.




### 1.3.3 Switching

- Switching is used to **connect networks**.
- A switch forwards data from one network to another when needed.
- Two main types of switched networks:
    1. Circuit-switched networks
    2. Packet-switched networks

**Circuit-Switched Network**

- A **dedicated communication path (circuit)** is established between two devices.
- The circuit remains reserved for the entire communication.
- Switches do not store data; they only forward it.
- Efficient only when fully utilized.
- Inefficient when traffic is low.

**Example:** Traditional telephone networks

Figure 1.12 *A heterogeneous network made of four WANs and three LANs*

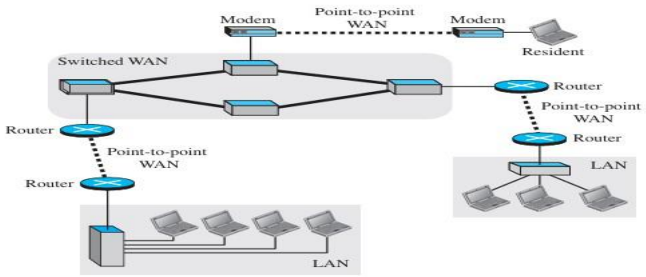
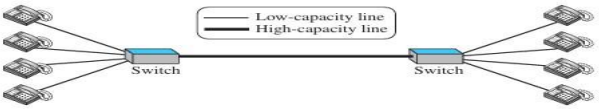



Figure 1.13 *A circuit-switched network*

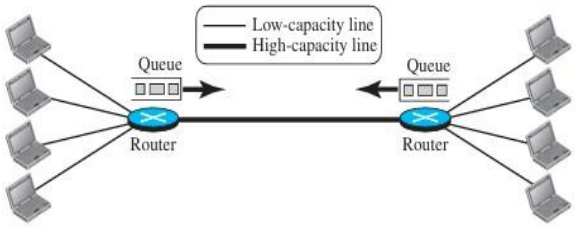



**Packet-Switched Network**

- Data is divided into **small units called packets**.
- Each packet is sent independently.
- Switches (routers) can **store and forward packets**.
- More efficient use of network capacity.
- Some delay may occur due to queuing.

**Example:** Computer networks and the Internet

Figure 1.14 *A packet-switched network*




### 1.3.4 The Internet

- The **Internet** (capital I) is the **largest internetwork**.
- It consists of **thousands of interconnected networks**.

- Organized into three levels:

    1. **Backbone networks** – large international ISPs

    2. **Provider networks** – national or regional ISPs

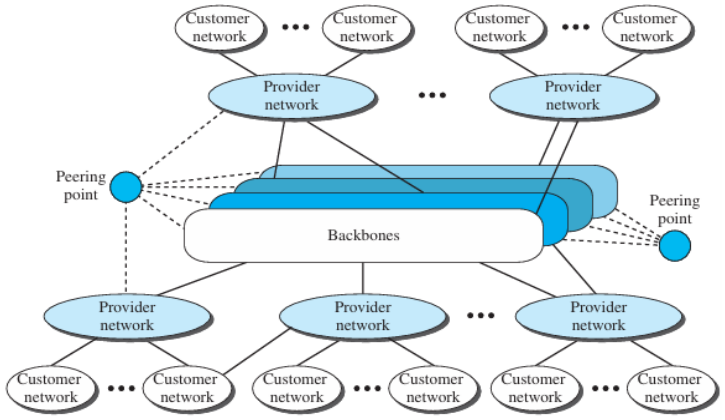
    3. **Customer networks** – end-user networks

- Networks are connected at **peering points**. At the top level, the backbones are large networks owned by some communication companies such as Sprint, Verizon (MCI), AT&T, and NTT. The back-bone networks are connected through some complex switching systems, called peering points.

- At the second level, there are smaller networks, called provider networks, that use the services of the backbones for a fee. The provider networks are connected to backbones and sometimes to other provider networks. The customer networks are networks at the edge of the Internet that actually use the services provided by the Internet. They pay fees to provider networks for receiving services.

- Backbones and providers together are called **ISPs (Internet Service Providers)**.'

**Figure 1.15** *The Internet today*




### 1.3.5 Accessing the Internet

To access the Internet, a user must be **physically connected to an ISP**, usually through a **point-to-point WAN**.

**Methods of Internet Access**

**1. Using Telephone Networks**

- **Dial-up Service**
    - Uses a modem
    - Very slow
    - Cannot use voice and Internet simultaneously
- **DSL Service**
    - High-speed
    - Allows voice and data at the same time

**2. Using Cable Networks**

- Uses cable TV infrastructure
- Provides higher speed
- Speed depends on number of users sharing the cable

**3. Using Wireless Networks**

- Uses wireless WANs
- Popular for homes and small businesses
- Provides mobility and flexibility

**4. Direct Connection**

- Used by large organizations
- Leases high-speed WAN
- Connects directly to a regional ISP

- Example: Universities, large corporations

## 1.4 Protocol Layering

A **protocol** is a set of rules that governs communication between devices.
In data communication and networking, **all communicating devices and intermediate devices must follow these rules** to communicate successfully.

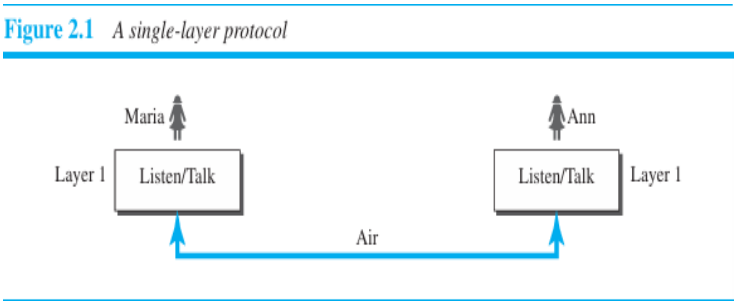
- When communication is **simple**, a single protocol is sufficient.
- When communication is **complex**, the task is divided into **multiple layers**.
- Each layer performs a specific function and uses its **own protocol**.
- This concept is called **protocol layering**

### 1.4.1 Scenarios

Let us develop two simple scenarios to better understand the need for protocol layering

**First Scenario: Single-Layer Communication**

- Maria and Ann are neighbors and communicate **face-to-face**.
- Communication occurs in **one layer**.
- They speak the same language and follow simple rules:
    - Greeting each other
    - Using appropriate words
    - Listening when the other speaks
    - Maintaining a two-way conversation
    - Saying goodbye politely
- This shows that **even simple communication needs rules (protocols)**.
- Different situations require **different protocols** (e.g., professor–student communication)




**Second Scenario: Multi-Layer Communication**

- Ann moves to a distant city.
- Maria and Ann communicate using **postal mail**.
- They want privacy, so they use **encryption and decryption**.
- Communication now occurs in **three layers**:

**Layer Functions:**

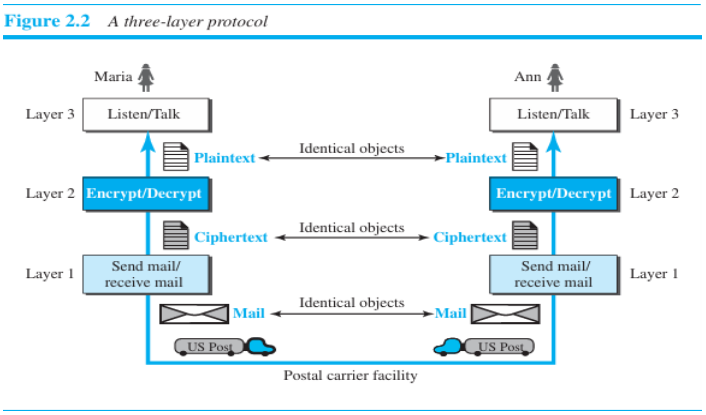
1. **Layer 3 (Application Layer)**
    - Talking and listening
    - Creates **plaintext** message
2. **Layer 2 (Security Layer)**
    - Encrypts plaintext into **ciphertext**
    - Decrypts ciphertext back into plaintext
3. **Layer 1 (Physical/Transport Layer)**
    - Sends and receives mail through the post office

**Working:**

- Sender:

- Plaintext → Encrypted → Sent as mail

- Receiver:
  - Mail → Decrypted → Plaintext → Read




**Need for Protocol Layering**

Protocol layering divides a **complex task into smaller, simpler tasks**.

**Advantages of Protocol Layering**

**1. Modularity**

- Each layer works **independently**.
- Changes in one layer do not affect other layers.
- Example: Only the encryption layer needs to be replaced if security is weak.

**2. Separation of Services and Implementation**

- A layer provides services without exposing how it works internally.
- Different implementations can provide the same service.

**3. Cost and Complexity Reduction**

- Intermediate devices need only some layers.

- Avoids making every device as complex as end systems.

**Disadvantage of Protocol Layering**

- Using multiple layers increases **overhead**.
- A single-layer system may seem simpler.
- However, changing or upgrading becomes difficult in single-layer systems

### 1.4.2 Principles of Protocol Layering First Principle: Bidirectional Capability

- Each layer must perform **two opposite tasks**, one in each direction.
- Examples:
    - Talk ↔ Listen
    - Encrypt ↔ Decrypt
    - Send ↔ Receive

**Second Principle: Identical Objects**

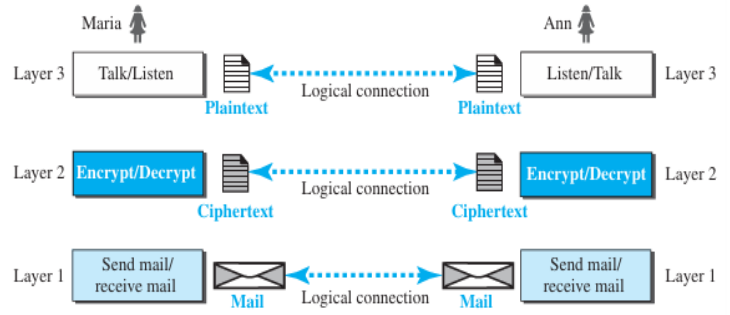
- The **same type of data object** must be exchanged at the same layer on both sides.
- Examples:
    - Layer 3: Plaintext
    - Layer 2: Ciphertext
    - Layer 1: Mail

### 1.4.3 Logical Connections

- Each layer communicates **logically** with the corresponding layer at the other end.

- This is called **peer-to-peer (layer-to-layer) communication**.
- The connection is **logical (imaginary)**, not physical.
- Logical connections help us understand how data flows through layers.

**Figure 2.3** *Logical connection between peer layers*




## 1.5 TCP/IP PROTOCOL SUITE

The **TCP/IP (Transmission Control Protocol / Internet Protocol)** is a **protocol suite** used in the Internet. A protocol suite is a **set of protocols organized in layers**, where each layer performs a specific function.

- TCP/IP is a **hierarchical layered model**.
- Each upper layer depends on the services provided by the lower layer.
- Initially, TCP/IP had **4 layers**, but today it is commonly represented as a **5-layer model**

**TCP/IP Layers (5-Layer Model)**

1. Application Layer
2. Transport Layer
3. Network Layer
4. Data-Link Layer
5. Physical Layer

### 1.5.1 Layered Architecture

In TCP/IP communication:

- **Hosts (source and destination)** use **all five layers**.
- **Routers** use only:
    - Network layer
    - Data-link layer
    - Physical layer
- **Link-layer switches** use only:
    - Data-link layer
    - Physical layer

**Key Points**

- Each link may use **different data-link and physical protocols**.
- Routers translate packets from one link protocol to another.
- Switches work only within a single link.

Let us assume that computer A communicates with computer B. As the figure shows, we have five communicating devices in this communication: source host (computer A), the link-layer switch in link 1, the router, the link-layer switch in link 2, and the destination host (computer B). Each device is involved with a set of layers depending on the role of the device in the internet. The two hosts are involved in all five layers; the source host needs to create a message in the application layer and send it down the layers so that it is physically sent to the destination host. The destination host needs to receive the communication at the physical layer and then deliver it through the other layers to the application layer

The router is involved in only three layers; there is no transport or application layer in a router as long as the router is used only for routing. Although a router is always involved in one network layer, it is involved in n combinations of link and physical layers in which n is the number of links the router is connected to. The reason is that each link may use its own data-link or physical protocol. For example, in the above figure, the router is involved in three links, but the message sent from source A to destination B is involved in two links.

Each link may be using different link-layer and physical-layer protocols; the router needs to receive a packet from link 1 based on one pair of proto cols and deliver it to link 2 based on another pair of protocols. A link-layer switch in a link, however, is involved only in two layers, data-link and physical. Although each switch in the above figure has two different connections, the connections are in the same link, which uses only one set of protocols. This means that, unlike a router, a link-layer switch is involved only in one data-link and one physical layer.

**Figure 2.4** *Layers in the TCP/IP protocol suite*

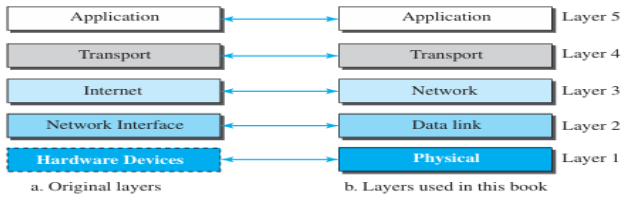
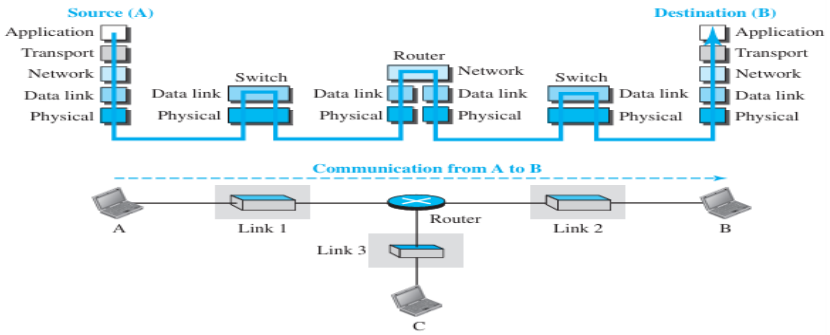



**Figure 2.5** *Communication through an internet*




### 1.5.2 Layers in the TCP/IP Protocol Suite

- **Application, Transport, and Network layers** perform **end-to-end communication**.
- **Data-link and Physical layers** perform **hop-to-hop communication**.
- Routers do not change:
  - Application data
  - Transport data
- Routers may modify:
  - Network-layer packets (e.g., fragmentation)

**Identical Objects at Each Layer**

- Application → Message
- Transport → Segment / User Datagram
- Network → Datagram
- Data-link → Frame
- Physical → Bit

**Figure 2.6** *Logical connections between layers of the TCP/IP protocol suite*

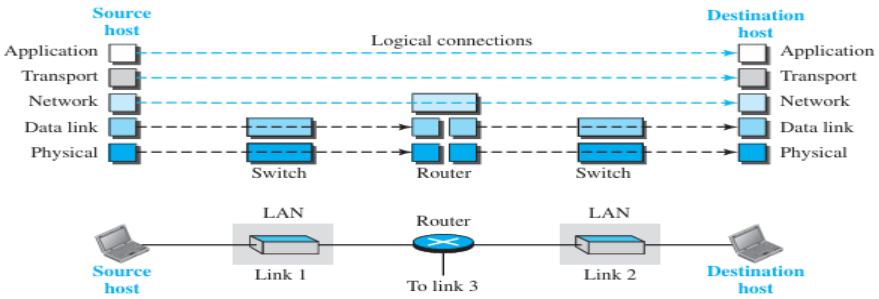
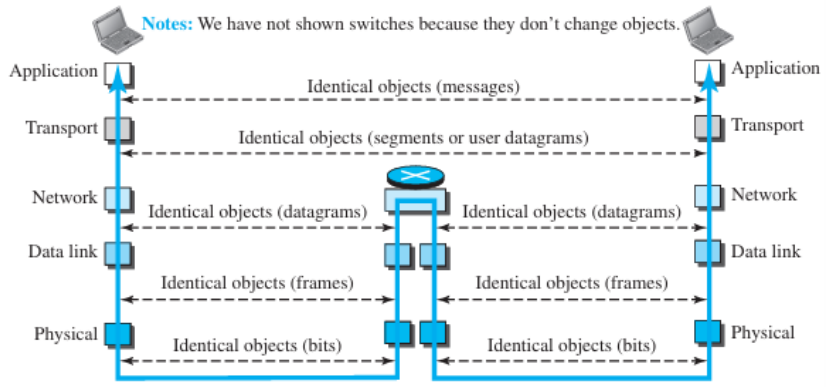



**Figure 2.7** *Identical objects in the TCP/IP protocol suite*




### 1.5.3 Description of Each Layer

### 1. Physical Layer

- Responsible for **transmitting bits** over the transmission medium.
- Converts bits into **electrical, optical, or radio signals**.

- Medium can be:
    - Cable
    - Optical fiber
    - Air (wireless)

**2. Data-Link Layer**

- Responsible for **moving packets across a single link**.
- Encapsulates network-layer datagrams into **frames**.
- Handles:
    - Framing
    - Error detection (and sometimes correction)
- TCP/IP does not define a specific data-link protocol.
- Supports all standard and proprietary protocols.

**3. Network Layer**

- Responsible for **host-to-host communication**.
- Selects the **best route** for packets using routers.
- Main protocol: **Internet Protocol (IP)**

**Functions of IP:**

- Packet formatting (datagram)
- Logical addressing
- Routing packets across networks

**Characteristics of IP:**

- Connectionless
- No flow control
- No error control
- No congestion control

**Supporting Protocols:**

- **ARP** – Finds link-layer address
- **ICMP** – Error reporting
- **IGMP** – Multicast management
- **DHCP** – Assigns IP addresses

**4. Transport Layer**

- Responsible for **process-to-process communication**.
- Delivers data from one application to the correct application on another host.

**Main Transport Protocols:**

- **TCP**
    - Connection-oriented
    - Reliable
    - Provides:
        - Flow control
        - Error control

        - Congestion control

- **UDP**
    - Connectionless
    - Unreliable
    - No flow, error, or congestion control
    - Low overhead and faster
- **SCTP**
    - Designed for modern multimedia applications

**5. Application Layer**

- Responsible for **process-to-process communication**.
- Provides services directly to the user.

**Common Application Protocols:**

- **HTTP** – Web access
- **SMTP** – Email transfer
- **FTP** – File transfer
- **TELNET / SSH** – Remote login
- **DNS** – Name to IP address mapping
- **SNMP** – Network management

### 1.5.4 Encapsulation and Decapsulation

**Encapsulation (Source Host)**

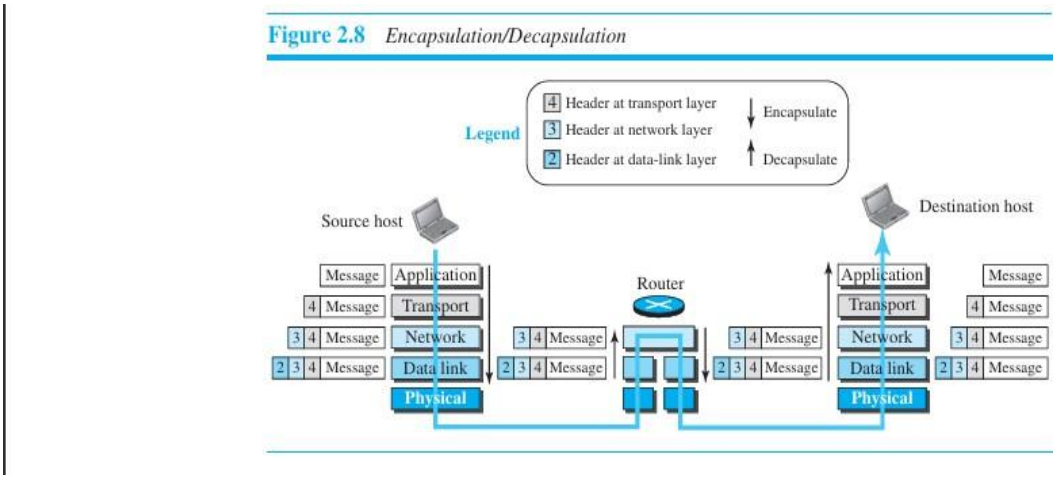
1. Application creates a **message**.
2. Transport layer adds header → **segment / user datagram**.
3. Network layer adds header → **datagram**.
4. Data-link layer adds header → **frame**.
5. Physical layer sends **bits**.

**At the Router**

- Frame is decapsulated.
- Datagram is examined.
- Datagram is re-encapsulated for the next link.

**Decapsulation (Destination Host)**

- Headers are removed layer by layer.
- Final message is delivered to the application layer.
- Error checking is performed during decapsulation.



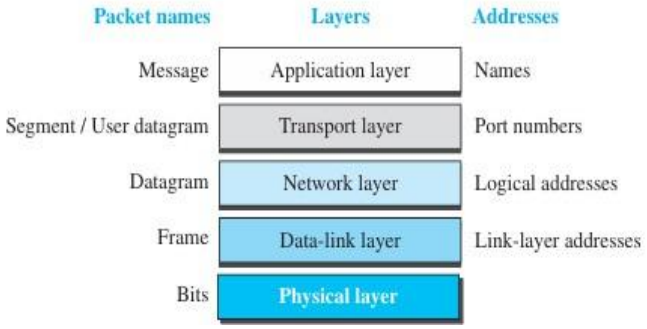

### 1.5.5 Addressing in TCP/IP

Each layer (except physical) uses its own addressing method:

we have logical communication between pairs of layers in this model. Any communication that involves two parties needs two addresses: source address and destination address.
Although it looks as if we need five pairs of addresses, one pair per layer, we normally have only four because the physical layer does not need addresses; the unit of data exchange at the physical layer is a bit, which defi nitely cannot have an address. Figure 2.9 shows the addressing at each layer. As the figure shows, there is a relationship between the layer, the address used in that layer, and the packet name at that layer. At the application layer, we normally use names to define the site that provides services, such as someorg.com, or the e-mail address, such as somebody@coldmail.com. At the transport layer, addresses are called port numbers, and these define the application-layer programs at the source and destination. Port numbers are local addresses that distinguish between several programs running at the same time. At the network-layer, the addresses are global, with the whole Internet as the scope. A network-layer address uniquely defines the connection of a device to the Internet.
The link-layer addresses, sometimes called MAC addresses, are locally defined addresses, each of which defines a specific host or router in a network (LAN or WAN).

| Layer | Address Type | Example |
|---|---|---|
| Application | Names | www.example.com |
| Transport | Port Numbers | 80, 443 |
| Network | IP Address | 192.168.1.1 |
| Data-Link | MAC Address | 00:1A:2B:3C |
| Physical | No address | Bits |

**Figure 2.9** *Addressing in the TCP/IP protocol suite*



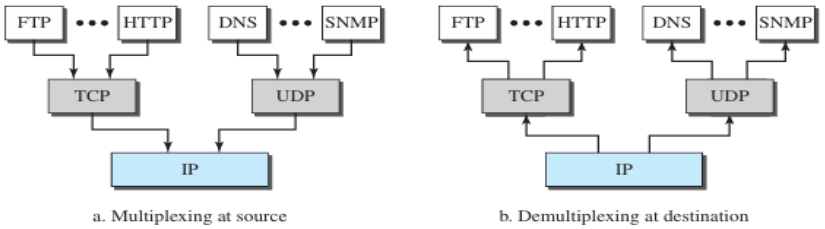

### 1.5.6 Multiplexing and Demultiplexing Multiplexing (Source)

- A protocol accepts data from **multiple higher-layer protocols**.
- Example:
    - TCP accepts data from HTTP, FTP, SMTP, etc.

**Demultiplexing (Destination)**

- A protocol delivers data to the **correct higher-layer protocol**.
- Identified using **header fields** (e.g., port numbers).

**Figure 2.10** *Multiplexing and demultiplexing*




## 1.6 THE OSI MODEL

The **International Organization for Standardization (ISO)** was established in **1947**.

ISO is a **multinational standards organization** with participation from most countries.

ISO introduced the **Open Systems Interconnection (OSI) model** in the **late 1970s**.

**ISO is the organization**, and **OSI is the model**.

**What is an Open System?**

- An **open system** allows communication between **different computer systems**, regardless of:
    - Hardware
    - Operating system
    - Internal architecture
- The goal is **interoperability** without changing existing systems.

**Purpose of the OSI Model**

- To provide a **standard framework** for network communication.
- To help designers understand how data flows in a network.
- To allow **flexible, robust, and interoperable network design**.
- The OSI model is **not a protocol**.
- It is a **reference model** used to design and understand protocols.

## OSI Layered Architecture

The OSI model consists of **seven layers**, each responsible for a specific task in data communication.

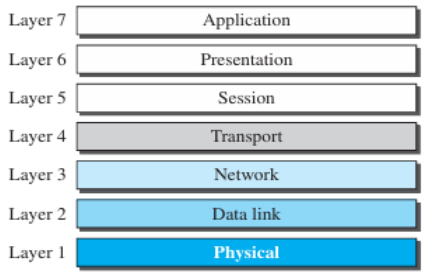
**The Seven OSI Layers (Top to Bottom)**

1. **Application Layer (Layer 7)**

- Provides services to the user
- Examples: file transfer, email, web access

2. **Presentation Layer (Layer 6)**
   - Data formatting
   - Encryption and decryption
   - Data compression
3. **Session Layer (Layer 5)**
   - Establishes, manages, and terminates sessions
   - Controls dialog between applications
4. **Transport Layer (Layer 4)**
   - End-to-end delivery
   - Error control and flow control
5. **Network Layer (Layer 3)**
   - Logical addressing
   - Routing and path selection
6. **Data Link Layer (Layer 2)**
   - Framing
   - Error detection
   - MAC addressing
7. **Physical Layer (Layer 1)**
   - Transmission of raw bits
   - Defines cables, signals, voltages

**Figure 2.11** *The OSI model*



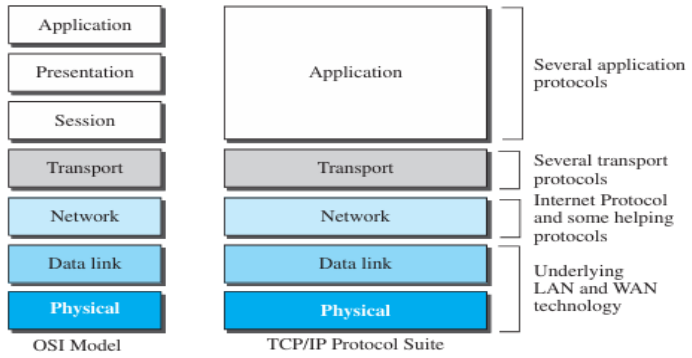

### 1.6.1 OSI Model versus TCP/IP Model

- TCP/IP has **5 layers**, while OSI has **7 layers**.
- The **Session and Presentation layers** are **missing in TCP/IP**.
- In TCP/IP:
    - These two layers are combined into the **Application layer**.

**Why TCP/IP Does Not Have Session and Presentation Layers?**

1. TCP/IP has **multiple transport protocols**.
    - Some session-related functions are handled by transport protocols.
2. TCP/IP applications are **flexible**.
    - If presentation or session features are needed, they are built into the application itself.

Figure 2.12 *TCP/IP and OSI model*




### 1.6.2 Lack of OSI Model's Success

Although the OSI model was well designed, it did **not replace TCP/IP**. The main reasons are:

**1. Late Arrival**

- TCP/IP was already **widely deployed**.
- Large investments had already been made.
- Replacing TCP/IP would have been **very costly**.

**2. Incomplete Layer Definitions**

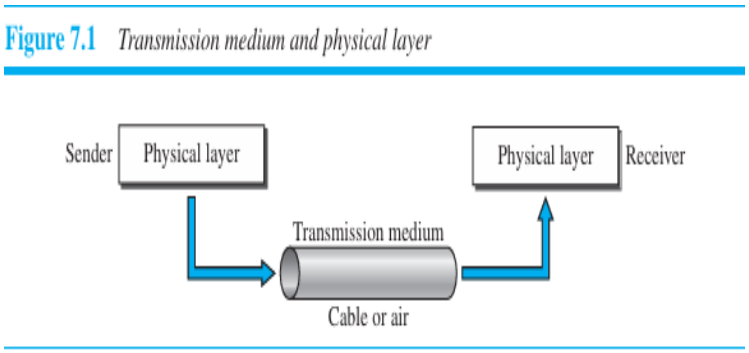
- Session and Presentation layers were **not fully defined**.
- Protocols and software for these layers were **incomplete**.

**3. Lower Performance**

- OSI implementations did not provide **better performance** than TCP/IP.
- Internet authorities saw **no strong reason** to switch

### 1.7 INTRODUCTION – TRANSMISSION MEDIA

- **Transmission media** are the physical paths through which data travels from a **sender to a receiver**.
- Transmission media are located **below the physical layer** and are directly controlled by it.
- Hence, transmission media are sometimes referred to as **Layer 0**.




**Definition of Transmission Medium**

- A **transmission medium** is **anything that carries information** from a source to a destination.
- Examples:
    - Air for human conversation
    - Smoke signals or semaphore
    - Mail carrier, truck, or airplane for written messages

**Transmission Media in Data Communication**

- In data communication, transmission media are more specific.
- Common transmission media include:
    - **Free space (air)**
    - **Metallic cables**

    - **Fiber-optic cables**

- Information is transmitted as **signals**, which are created by converting data into another form.

**Historical Development of Transmission Media**

- **Telegraph (19th century)**
    - Invented by Morse
    - Used electric signals over metallic wires
    - Slow and limited
- **Telephone (1869)**
    - Enabled long-distance voice communication
    - Used metallic wires
    - Communication was unreliable due to poor wire quality
- **Wireless Communication (1895)**
    - Hertz transmitted high-frequency signals
    - Marconi sent wireless telegraph messages across the Atlantic

**Modern Communication Media**

- Improved **metallic cables** such as:
    - Twisted-pair cables
    - Coaxial cables
- **Optical fiber** greatly increased data transmission speed.
- **Free space (air, vacuum, water)** is used efficiently using:

- Modulation
- Multiplexing techniques

**Signals and Electromagnetic Energy**

- Computers and communication devices use **signals to represent data**.
- Signals travel as **electromagnetic energy** through transmission media.
- Electromagnetic energy consists of:
  - Electric and magnetic fields vibrating together
- The electromagnetic spectrum includes:
  - Radio waves
  - Infrared
  - Visible light
  - Ultraviolet
  - X-rays, gamma rays, and cosmic rays
- Only **certain portions** of the spectrum are used for communication.

**Classification of Transmission Media**

Transmission media are classified into **two broad categories**:

**1. Guided Media**

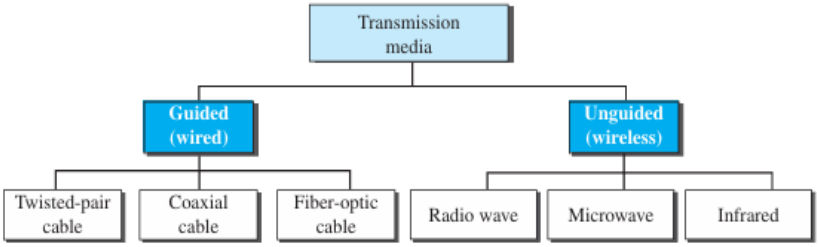
- Signals travel through a **physical path**.
  - Examples:
  - Twisted-pair cable

- Coaxial cable
- Fiber-optic cable

**2. Unguided Media**

- Signals travel through **free space**.
- Example:
  - Wireless communication (air)

**Figure 7.2** *Classes of transmission media*




## 1.8 GUIDED MEDIA

- **Guided media** provide a **physical path (conduit)** for signals to travel from one device to another.
- In guided media, signals are **directed and contained** within the physical boundaries of the medium.
- Types of guided media:
  1. **Twisted-pair cable**
  2. **Coaxial cable**
  3. **Fiber-optic cable**
- **Twisted-pair and coaxial cables** use **metallic (copper) conductors** and carry signals as **electric current**.

- **Fiber-optic cables** carry signals as **light**.

### 1.8.1 Twisted-Pair Cable

- A **twisted-pair cable** consists of **two insulated copper wires** twisted together.
- One wire carries the **signal**, and the other acts as a **ground/reference**.
- The receiver detects data by measuring the **difference between the two wires**.

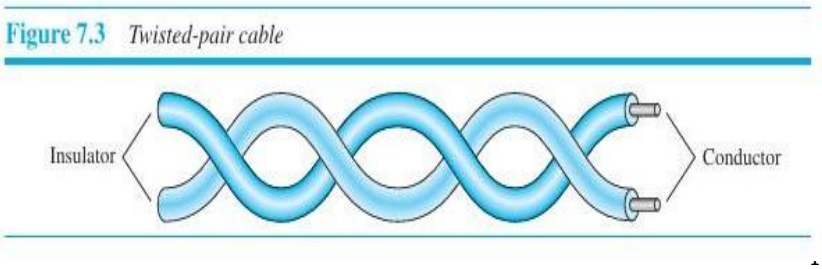


Figure 7.3 *Twisted-pair cable*

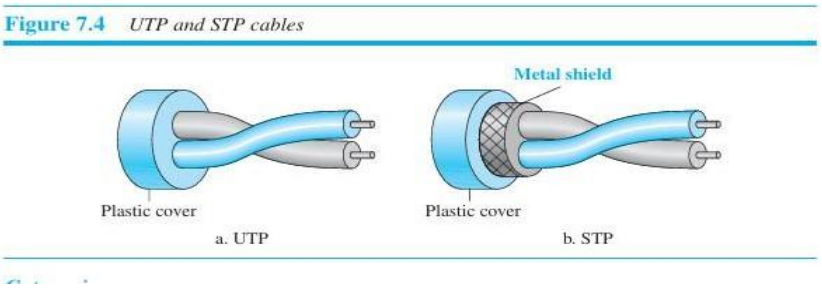
**Noise and Crosstalk Reduction**

- External interference such as **noise and crosstalk** can affect both wires.
- If the wires are **parallel**, interference affects them differently, causing errors.
- **Twisting the wires** balances the interference:
    - One wire is closer to the noise source in one twist.
    - In the next twist, the other wire is closer.
- This causes unwanted signals to **cancel each other out** at the receiver.
- **More twists per unit length** result in **better signal quality**.

**Unshielded vs Shielded Twisted-Pair Unshielded Twisted-Pair (UTP)**

- Most commonly used twisted-pair cable.

- No metallic shield.
- Lightweight, flexible, and inexpensive.
- Widely used in LANs and telephone lines.




**Shielded Twisted-Pair (STP)**

- Has a **metal foil or braided shield** around the wire pairs.
- Provides better protection against noise and crosstalk.
- Bulkier and more expensive.
- Rarely used outside specialized environments (e.g., IBM systems).

**Categories of UTP Cables (EIA Standards)**

UTP cables are classified by the **Electronic Industries Association (EIA)** based on performance.

| Category | Data Rate | Typical Use |
|---|---|---|
| CAT 1 | < 0.1 Mbps | Telephone |
| CAT 2 | 2 Mbps | T-lines |

| | | |
|---|---|---|
| CAT 3 | 10 Mbps | LANs |
| CAT 4 | 20 Mbps | LANs |
| CAT 5 | 100 Mbps | LANs |
| CAT 5e | 125 Mbps | LANs (reduced crosstalk) |
| CAT 6 | 200 Mbps | High-speed LANs |
| CAT 7 | 600 Mbps | High-performance LANs |

**Connectors**

- The most common connector used with UTP is **RJ-45 (Registered Jack-45)**.
- It is a **keyed connector**, meaning it can be inserted in only one direction.
- Widely used in Ethernet networks.

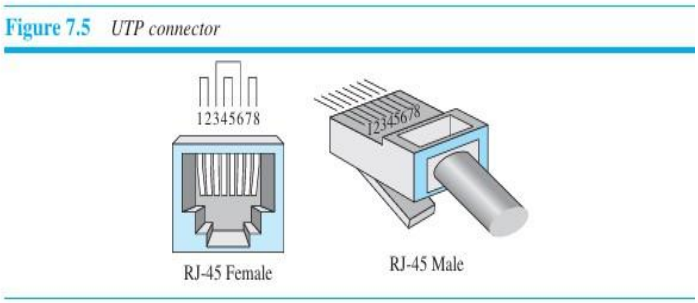


Figure 7.5 *UTP connector*

**Performance of Twisted-Pair Cable**

- Performance is affected by:

- **Frequency**
- **Distance**
- **Wire thickness (gauge)**

- As **frequency increases**, **attenuation (signal loss)** increases sharply.
- Thicker wires (lower gauge number) have **less attenuation**.

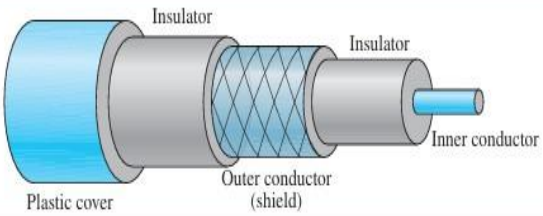
**Applications of Twisted-Pair Cable**

- **Telephone networks**
  - Used in the **local loop** between subscriber and telephone exchange.
- **DSL (Digital Subscriber Line)**
  - Uses twisted-pair cables for high-speed Internet access.
- **Local Area Networks (LANs)**
  - Ethernet technologies such as **10Base-T** and **100Base-T**.

**1.8.2 Coaxial Cable**

- **Coaxial cable (coax)** carries signals of **higher frequency** than twisted-pair cable.
- It has **better shielding**, which reduces noise and interference.
- Widely used in **cable TV** and **early Ethernet networks**.

Figure 7.7 *Coaxial cable*




**Structure of Coaxial Cable**

A coaxial cable consists of **four layers**:

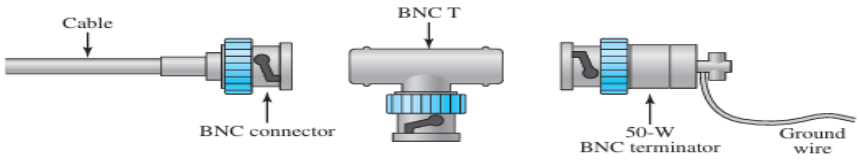
1. **Inner conductor**
    - Solid or stranded copper wire
    - Carries the actual signal
2. **Inner insulator (dielectric)**
    - Separates inner conductor from outer conductor
3. **Outer conductor (shield)**
    - Metal foil, braid, or both
    - Acts as:
        - Shield against noise
        - Second conductor to complete the circuit
4. **Outer plastic cover**
    - Protects the cable from physical damage

**Coaxial Cable Standards (RG Ratings)**

- Coaxial cables are classified using **RG (Radio Government) ratings**.

- Each RG number defines:
    - Wire gauge
    - Type and thickness of insulation
    - Shield construction
    - Outer casing

**Figure 7.8** *BNC connectors*




| RG Type | Impedance | Use |
|---|---|---|
| RG-59 | 75 Ω | Cable TV |
| RG-58 | 50 Ω | Thin Ethernet |
| RG-11 | 50 Ω | Thick Ethernet |

**Coaxial Cable Connectors**

The most commonly used connectors are **BNC (Bayonet Neill-Concelman)** connectors.

**Types of BNC Connectors**

- **BNC Connector**
    - Connects cable end to devices (e.g., TV)
- **BNC T Connector**

- Used in Ethernet to connect devices to the cable

- **BNC Terminator**
  - Placed at cable ends
  - Prevents signal reflection

**Performance of Coaxial Cable**

- Coaxial cable has:
  - **Higher bandwidth** than twisted-pair cable
  - **Higher attenuation** (signal loss over distance)
- Because attenuation increases rapidly:
  - **Repeaters** are required at regular intervals
- Suitable for **high-frequency transmission**, but not ideal for very long distances without amplification.

**Applications of Coaxial Cable**

**1. Telephone Networks (Earlier)**

- Used in analog systems to carry up to **10,000 voice signals**
- Used in digital systems for data rates up to **600 Mbps**
- Mostly replaced today by **fiber-optic cables**

**2. Cable TV Networks**

- Traditional cable TV networks used only coaxial cables
- Modern systems use **hybrid fiber-coaxial (HFC)** networks

- Coaxial cable is used near customer premises
- **RG-59** is commonly used

### 3. Ethernet LANs (Early)

- Used due to high bandwidth

#### 10Base-2 (Thin Ethernet)

  - Uses RG-58
  - 10 Mbps
  - Maximum length: **185 m**

- **10Base-5 (Thick Ethernet)**
  - Uses RG-11
  - 10 Mbps
  - Maximum length: **500 m**
  - Uses special connectors

**Advantages of Coaxial Cable**

- Better noise immunity than twisted-pair
- Higher bandwidth
- Suitable for high-frequency signals

**Disadvantages of Coaxial Cable**

- Higher attenuation
- Bulkier and less flexible

- Largely replaced by fiber optics in modern networks

### 1.8.3 Fiber-Optic Cable

- A **fiber-optic cable** is made of **glass or plastic** and carries data in the form of **light signals**.
- It works on the principle of **reflection of light**.
- Compared to copper cables, fiber optics provide **very high bandwidth** and **low signal loss**.

#### Nature of Light and Principle of Operation

- Light travels in a straight line in a **uniform medium**.
- When light moves from one medium to another with **different density**, it changes direction.
- Based on the **angle of incidence**:
    - Less than critical angle → **Refraction**
    - Equal to critical angle → Light travels along the boundary
    - Greater than critical angle → **Reflection**

#### Total Internal Reflection

- Optical fibers use **total internal reflection**.
- The fiber consists of:
    - **Core** (denser material)
    - **Cladding** (less dense material)

- Light entering the core is reflected continuously by the cladding and guided along the fiber.

**Figure 7.10** *Bending of light ray*

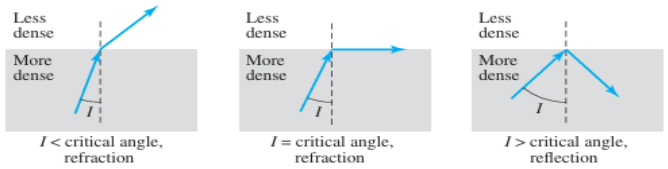
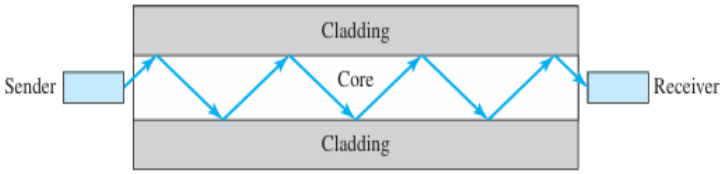



**Figure 7.11** *Optical fiber*




**Structure of Optical Fiber**

1. **Core**
    - Central glass/plastic region
    - Carries light signals
2. **Cladding**
    - Surrounds the core
    - Lower density than core
    - Causes reflection
3. **Protective layers**
    - Plastic coating
    - Kevlar strands for strength
    - Outer jacket (PVC or Teflon)

**Propagation Modes in Optical Fiber**

Optical fibers support **two main propagation modes**:

### 1. Multimode Fiber

- Multiple light rays travel through the core.
- Used for **short distances**.

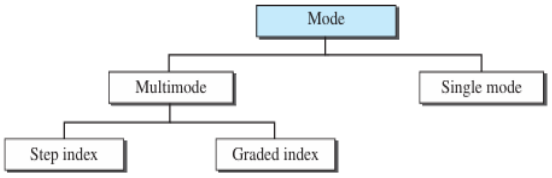
### a) Step-Index Multimode

- Core has **uniform density**.
- Light rays reflect sharply.
- Causes **more signal distortion**.

### b) Graded-Index Multimode

- Core density **gradually decreases** from center to edge.
- Light follows curved paths.
- **Less distortion** than step-index fiber.

**Figure 7.12** *Propagation modes*




### 2. Single-Mode Fiber

- Uses **single, narrow light beam**.
- Core diameter is **very small**.
- Light travels almost straight.

- **Negligible distortion and delay**.
- Used for **long-distance and high-speed communication**.

**Figure 7.13** *Modes*

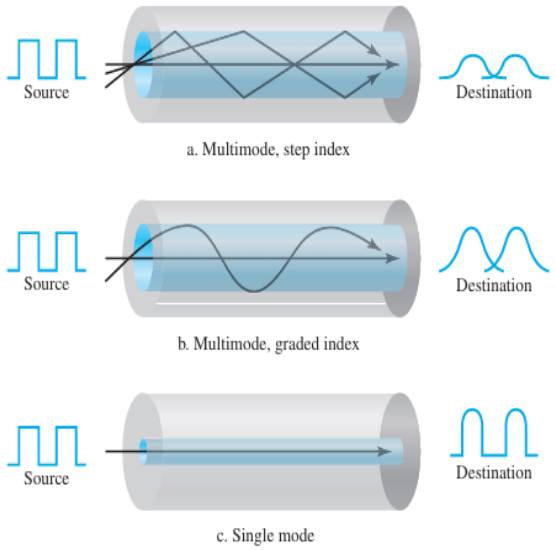
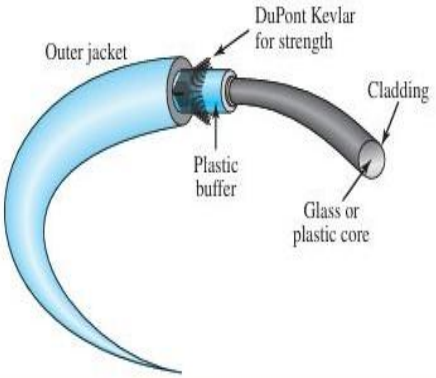



**Figure 7.14** *Fiber construction*




**Fiber Sizes**

Optical fibers are identified by **core/cladding diameter (µm)**:

| Type | Core/Cladding (µm) |
|---|---|

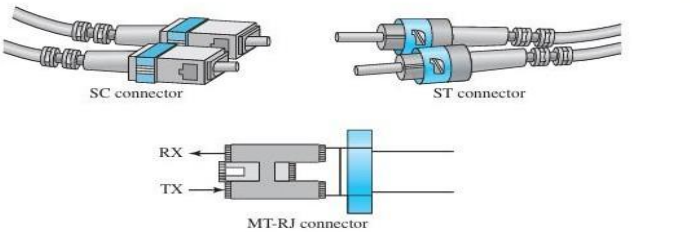
| | |
|---|---|
| Multimode | 50/125 |
| Multimode | 62.5/125 |
| Multimode | 100/125 |
| Single-mode | 7/125 |

**Fiber-Optic Cable Connectors**

Common connectors include:

- **SC (Subscriber Connector)**
  - Push–pull locking
  - Used in cable TV
- **ST (Straight Tip)**
  - Bayonet locking
  - Used in networking devices
- **MT-RJ**
  - Same size as RJ-45
  - Compact connector

**Figure 7.15** *Fiber-optic cable connectors*



**Performance of Fiber-Optic Cable**

- Fiber has **very low attenuation**.
- Attenuation remains almost **flat over wide wavelengths**.
- Requires **far fewer repeaters** compared to copper cables.
- Supports extremely high data rates.
- With **Wavelength Division Multiplexing (WDM)**, data rates up to **1600 Gbps** are possible.

**Applications of Fiber-Optic Cable**

- **Backbone networks** (e.g., SONET)
- **High-speed Internet**
- **Cable TV (Hybrid Fiber-Coaxial networks)**
- **LANs**
    - 100Base-FX (Fast Ethernet)
    - 1000Base-X (Gigabit Ethernet)

**Advantages of Fiber-Optic Cable**

- Very high bandwidth
- Low signal attenuation
- Immune to electromagnetic interference
- Resistant to corrosion
- Lightweight
- More secure against tapping

**Disadvantages of Fiber-Optic Cable**

- **High installation and maintenance cost**
- Requires **skilled technicians**
- **Unidirectional transmission**
    - Two fibers needed for full-duplex communication
- Not economical for **low-bandwidth requirements**

## 1.9 UNGUIDED MEDIA: WIRELESS

- **Unguided media** transmit electromagnetic waves **without using a physical conductor**.
- This type of communication is called **wireless communication**.
- Signals are transmitted through **free space (air, vacuum, or water)**.
- Wireless signals are **broadcast**, so they can be received by any device tuned to the same frequency.

**Electromagnetic Spectrum for Wireless Communication**

- Wireless communication uses frequencies from **3 kHz to 900 THz**.
- This range includes:
    - Radio waves
    - Microwaves
    - Infrared waves
- Not all frequencies are usable; many are **regulated by government authorities**.

**Propagation Methods of Wireless Signals**

Wireless signals can travel in **three different ways**:

**1. Ground Propagation**

- Used for **low-frequency signals (below 2 MHz)**.
- Waves follow the **curvature of the Earth**.
- Distance depends on **signal power**.
- Used for **radio navigation**.

**2. Sky Propagation**

- Used for **medium frequencies (2–30 MHz)**.
- Waves are sent upward and **reflected back by the ionosphere**.
- Allows **long-distance communication** with low power.
- Used in **AM radio** and long-distance radio communication.

**3. Line-of-Sight Propagation**

- Used for **high frequencies (above 30 MHz)**.
- Waves travel in **straight lines**.
- Requires **aligned antennas**.
- Affected by Earth's curvature and obstacles.
- Used in **microwave, satellite, and infrared communication**.

**Frequency Bands**

- The radio and microwave spectrum is divided into **bands**:
    - VLF, LF, MF, HF, VHF, UHF, SHF, EHF
- Each band has:

- A specific frequency range
- Defined propagation method
- Specific applications

- Usage is **regulated** by authorities.

**Classification of Wireless Transmission**

Wireless media are divided into **three main types**:

1. Radio Waves
2. Microwaves
3. Infrared Waves

**Figure 7.17** *Electromagnetic spectrum for wireless communication*

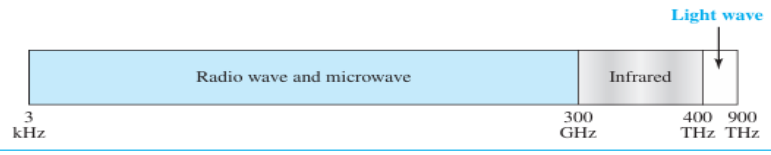
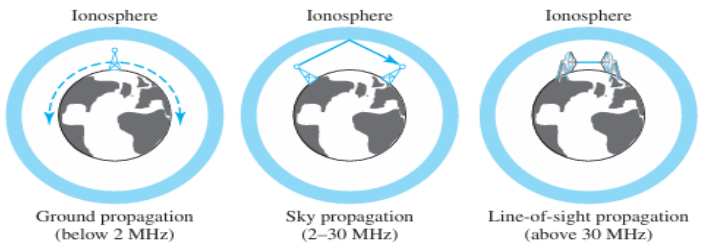



**Figure 7.18** *Propagation methods*




### 1.9.1 Radio Waves

- Frequency range: **3 kHz to 1 GHz**
- Mostly **omnidirectional**.
- Transmitted in **all directions**.

- Sending and receiving antennas **need not be aligned**.

**Characteristics**

- Can travel **long distances** (especially sky propagation).
- Can **penetrate walls**.
- Susceptible to **interference**.
- Lower bandwidth → **lower data rate**.
- Heavily **regulated**.

**Antenna Type**

- Uses **omnidirectional antennas**.

**Applications**

- AM and FM radio
- Television broadcasting
- Maritime radio
- Cordless phones
- Paging systems

**Radio waves are mainly used for multicast communication.**

**Figure 7.19** *Omnidirectional antenna*



### 1.9.2 Microwaves

- Frequency range: **1 GHz to 300 GHz**
- **Unidirectional**.
- Signals can be **narrowly focused**.

**Characteristics**

- **Line-of-sight propagation**
- Requires **aligned antennas**
- Cannot penetrate walls
- Wide bandwidth → **high data rate**
- Some frequency bands require permission

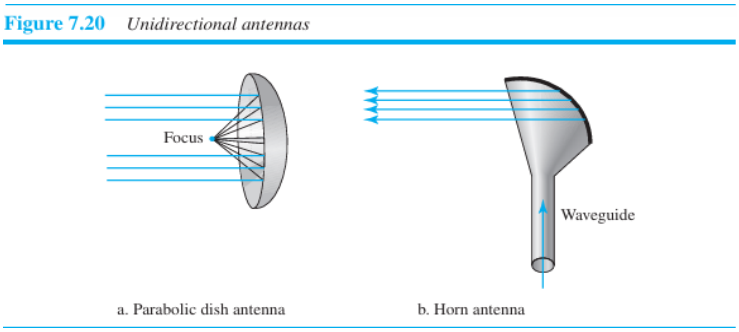
**Antenna Types**

- **Parabolic dish antenna-**A parabolic dish antenna is based on the geometry of a parabola: Every line parallel to the line of symmetry (line of sight) reflects off the curve at angles such that all the lines intersect in a common point called the focus. The parabolic dish works as a funnel, catching a wide range of waves and directing them to a common point. In this way, more of the signal is recovered than would be possible with a single-point receiver.
- **Horn antenna-**A horn antenna looks like a gigantic scoop. Outgoing transmissions are broadcast up a stem (resembling a handle) and deflected outward in a series of narrow parallel beams by the curved head. Received transmissions are collected by the scooped shape of the horn, in a manner similar to the parabolic dish, and are deflected down into the stem.

**Applications**

- Cellular phone networks
- Satellite communication
- Wireless LANs

**Microwaves are mainly used for unicast (one-to-one) communication.**




### 1.9.3 Infrared Waves

- Frequency range: **300 GHz to 400 THz**
- Used for **short-range communication**.
- Cannot penetrate walls.
- Strongly affected by **sunlight**.

### Characteristics

- No interference between adjacent rooms
- Cannot be used for long-distance communication
- Requires **line-of-sight**

### Applications

- Infrared remote controls
- Wireless keyboards and mice
- Device-to-device communication using **IrDA standards**

### IrDA (Infrared Data Association)

- Defines standards for infrared communication
- Initial data rate: **75 kbps**
- Current data rate: **up to 4 Mbps**
- Distance: up to **8 meters**

### 1.10 PACKET SWITCHING

- In **packet switching**, a message is divided into **small units called packets**.
- Each packet may be **fixed-size or variable-size**, depending on the network and protocol.
- Packets are sent **independently** through the network.
- **No resources are reserved** in advance.
- Bandwidth and processing time are **allocated on demand**.
- Packets are processed on a **first-come, first-served basis**.
- This may cause **delay**, similar to waiting without a reservation.

**In a packet-switched network, resources are allocated only when needed. Types of Packet-Switched Networks**

There are **two types**:

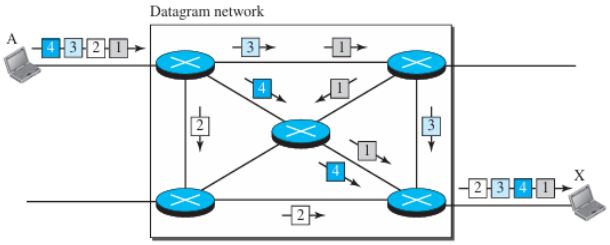
1. Datagram Networks
2. Virtual-Circuit Networks

### 1.10.1 Datagram Networks

- In a **datagram network**, each packet is treated **independently**.
- Packets are called **datagrams**.

- Even packets from the same message may:
    - Take **different paths**
    - Arrive **out of order**
    - Experience **different delays**
- Packets may be **lost** due to congestion.
- Reordering and retransmission are handled by **upper-layer protocols** (like TCP).

**Figure 8.7** *A datagram network with four switches (routers)*



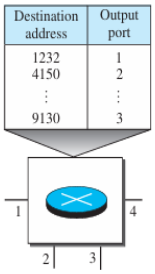

**Key Characteristics**

- **Connectionless network**
- **No setup or teardown phase**
- Switches do **not store connection state**
- Each packet is routed independently

**Routing in Datagram Networks**

- Each switch (router) has a **routing table**.
- The table is based on the **destination address**.
- Routing tables are **dynamic** and updated periodically.
- When a packet arrives:
    - Destination address is checked
    - Output port is selected from the routing table

**Figure 8.8** *Routing table in a datagram network*



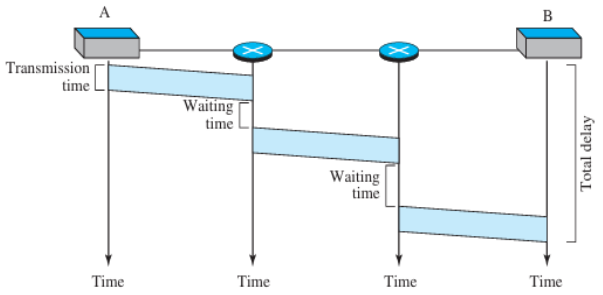

**Destination address remains the same throughout the journey. Efficiency of Datagram Networks**

- More efficient than circuit switching
- Resources are used **only when packets exist**
- Idle time can be used by other packets

**Delay in Datagram Networks**

- Delay is **variable**
- Each packet may experience:
    - Transmission delay
    - Propagation delay
    - Waiting (queuing) delay
- Packets of the same message may arrive with **unequal delays**

Figure 8.9 *Delay in a datagram network*

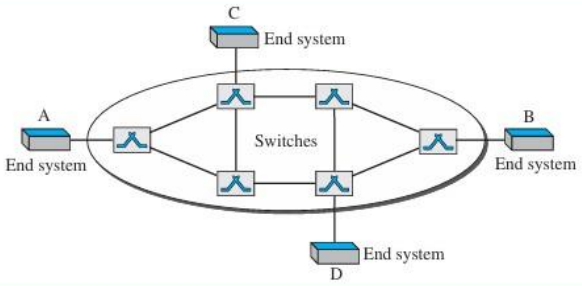



### 1.10.2 Virtual-Circuit Networks

A **virtual-circuit network** combines features of:

- Circuit-switched networks
- Datagram networks

Figure 8.10 *Virtual-circuit network*




**Characteristics**

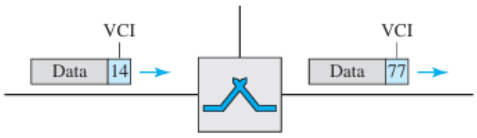
1. Has **setup, data transfer, and teardown phases**
2. Packets are **packetized**, not continuous
3. All packets follow the **same path**
4. Each packet carries a **virtual-circuit identifier (VCI)** instead of a full address
5. Implemented mainly at the **data-link layer**

**Addressing in Virtual-Circuit Networks**

**1. Global Address**

- Used **only during setup**
- Identifies source and destination globally

**Figure 8.11** *Virtual-circuit identifier*




**2. Virtual-Circuit Identifier (VCI)**

- Used during **data transfer**
- Has **local significance only**
- Changes at **each switch**
- Smaller than global addresses

**Three Phases of Virtual-Circuit Networks**

**1. Setup Phase**

- Source sends a **setup request**
- Switches create **table entries**
- Destination assigns a VCI
- Acknowledgment completes table entries

**2. Data Transfer Phase**

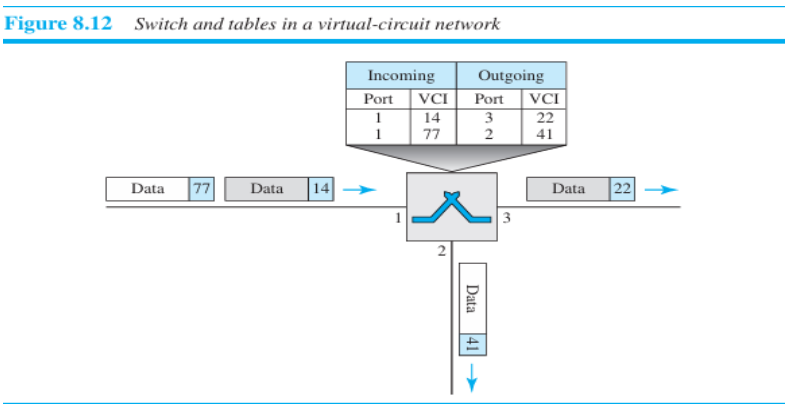
- Frames travel along the **predefined path**

- Each switch:
    - Reads incoming VCI
    - Replaces it with a new outgoing VCI
    - Forwards the frame

**A virtual circuit is logical, not physical.**

### 3. Teardown Phase

- Source sends a **teardown request**
- Destination confirms
- Switches **delete table entries**

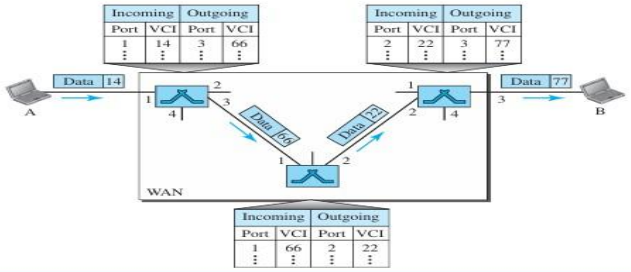



**Efficiency of Virtual-Circuit Networks**

- If resources are reserved during setup:
    - Constant delay
    - No waiting for packets
- If resources are allocated on demand:

- Advantage:
    - Availability can be checked before sending data

**Figure 8.13** *Source-to-destination data transfer in a virtual-circuit network*




***Setup Request***

A setup request frame is sent from the source to the destination. Figure 8.14 shows the process.

**Figure 8.14** *Setup request in a virtual-circuit network*

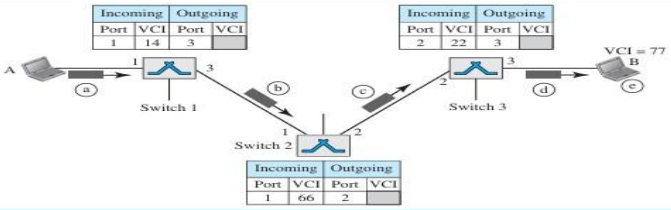
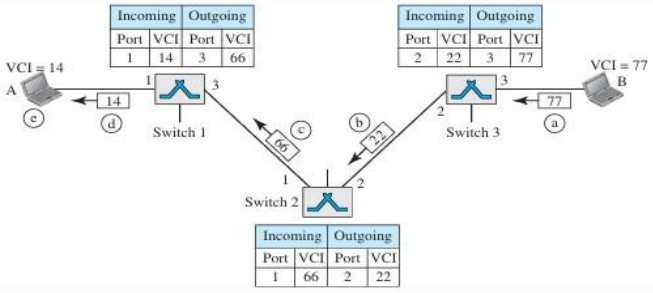



**Figure 8.15** *Setup acknowledgment in a virtual-circuit network*



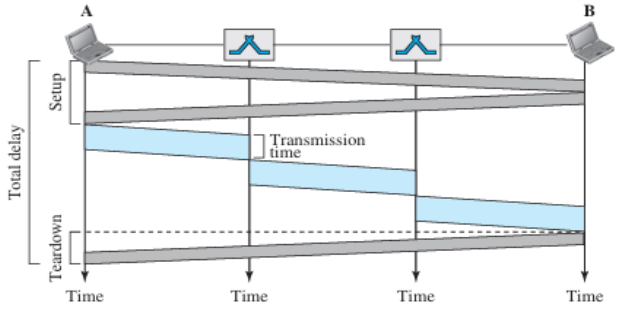

## Delay in Virtual-Circuit Networks

- Includes:
    - Setup delay
    - Transmission delay
    - Propagation delay

- Teardown delay

- After setup, packet delays are **more predictable** than datagram networks
- Total delay 1 3T 1 3τ 1 setup delay 1 teardown delay

**Figure 8.16** *Delay in a virtual-circuit network*




Circuit-Switched Technology in WANs

virtual-circuit networks are used in switched WANs such as ATM networks. The data-link layer of these technologies is well suited to virtual circuit technology. Switching at the data-link layer in a switched WAN is normally implemented by using virtual-circuit techniques.